Department of Computer Science and Engineering (CSE)
BRAC University

Practice Problem Set 5

CSE251 – Electronic Devices and Circuits

MOSFET Circuits

S-Model, SR-Model, Real MOSFET Model, Logic Function Implementation, Method of Assumed States, and Multistage Circuits

V_G, V_D, and V_S denote the gate, drain, and source voltages of MOSFETs, respectively . "Sat" and "Triode" refer to the saturation and triode regions, respectively. k_n and k_n' denote MOSFET's transconductance and process transconductance parameters, respectively. V_{T_n} denote the threshold voltage of the MOSFETs. I_{DS} denote the drain to source current.

Course Description, COs, and Policies

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Midterm and Final Questions

For any error or issues, contact purbayan.das@bracu.ac.bd

Problem 1

- Give a switch-MOSFET implementation of the following logic functions. A, B, C, D, E and F are the Boolean inputs and f is the Boolean output. Here, "+" and "." represent the logical OR and AND operations, respectively.

I. $f = A.\bar{B}$ [Spring 2024]

II. $f = \overline{A.B.C + D.E}$ [Summer 2024]

III. $f = A.B.(C + D)$ [Summer 2024]

IV. $f = A.B + \bar{A}.\bar{B}$ [Spring 2024]

V. $f = \overline{A.C} + \overline{B + C}$

VI. $f = (A.B + C).D$

VII. $f = A.B + B.C + C.A$ [Fall 2024]

VIII. $f = (A + B).(C + D)$

IX. $f = (A + B).C + D.E$ [Fall 2025]

X. $f = (A.B + C).(D + E)$

XI. $f = (A.B + C).D.(E + F)$

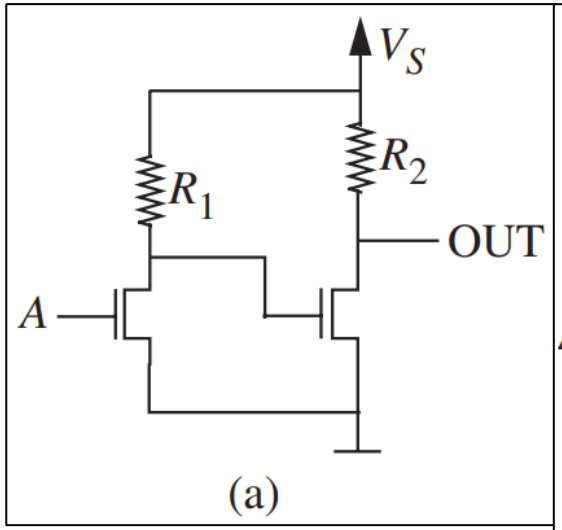
XII. $f = A \oplus B$

XIII. $f = \overline{(A + B + C).D + \overline{D}}$ [Summer 2025]

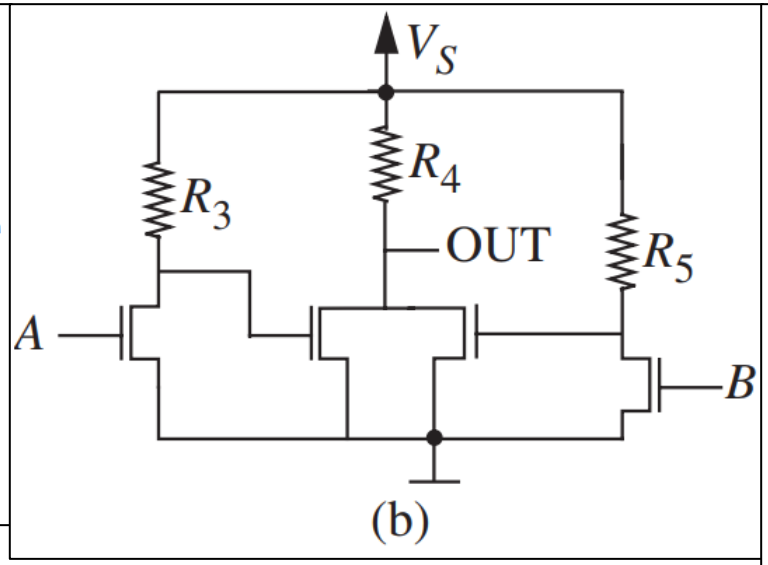
XIV. $f = \overline{C.(A + B)}.(A + \bar{B} + C)$

Problem 2

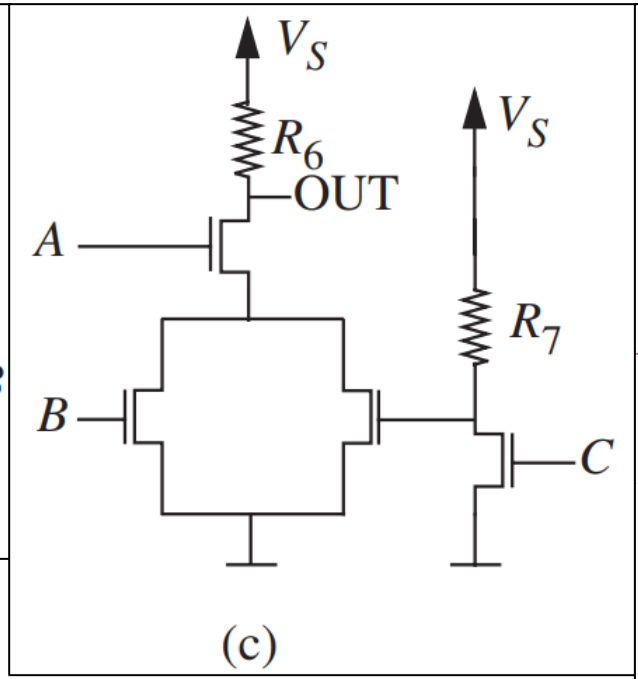
- Write a Boolean expression that describes the function of each of the circuits below.



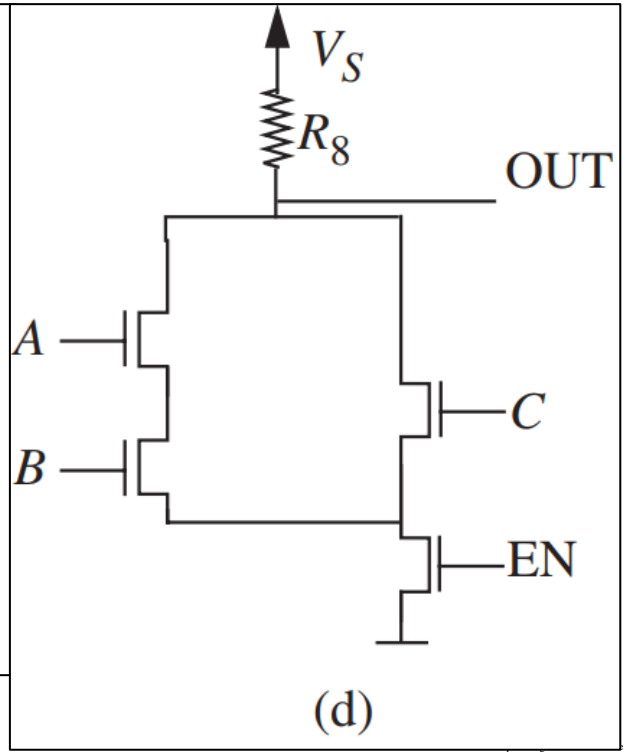
(a)



(b)



(c)



(d)

Problem 3

- Draw voltage transfer characteristics (VTC) for the following logic gates implemented using MOSFETs. Use S −Model.

  I. Inverter

  II. 2-input NAND Gate

  III. 2-input NOR Gate

  IV. 3-input AND Gate or $f = A.B.C$

  V. 3-input OR Gate or $f = A + B + C$

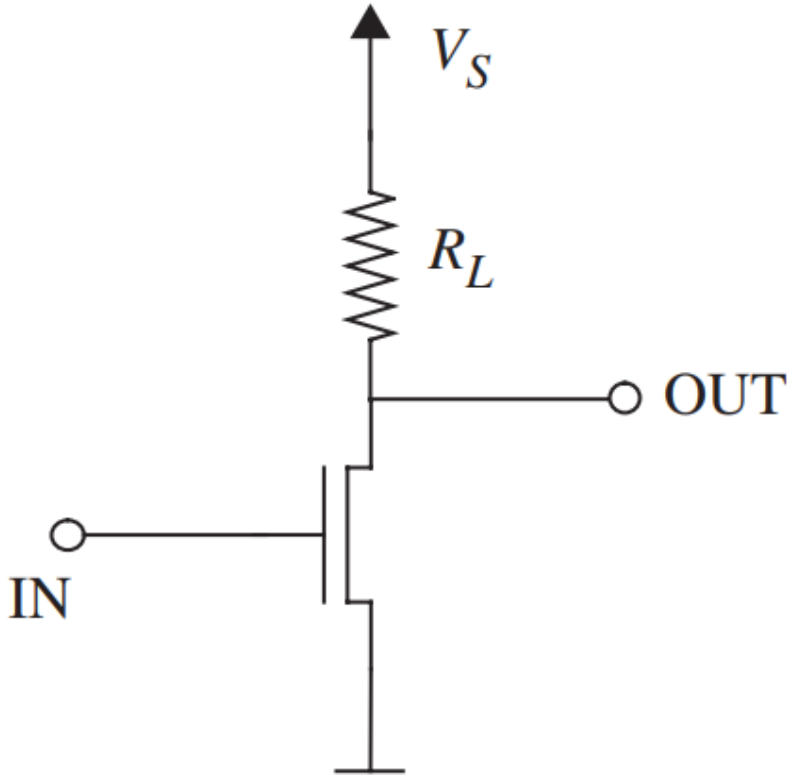
Problem 4

- The MOSFET in the following inverter circuit has a threshold voltage $V_{Tn} = 2\ V$ and $R_{ON} = 8\ k\Omega$. For the circuit, $V_S = 5\ V$. Draw the $OUT\ vs. IN$ (VTC) graph by modeling the MOSFETs using

  I. S −Model and

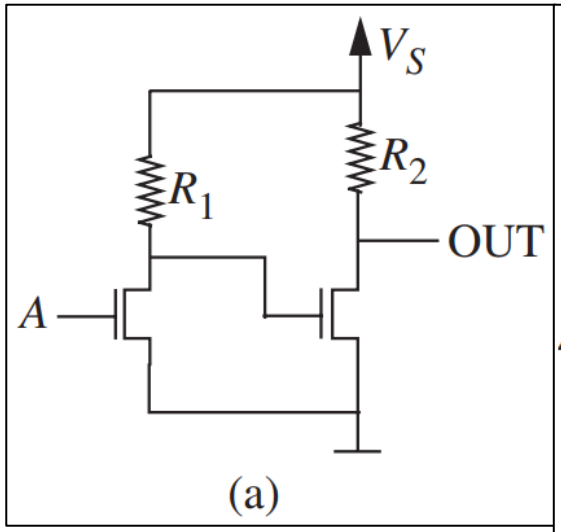
  II. SR −Model with $R_L = 10\ k\Omega$.

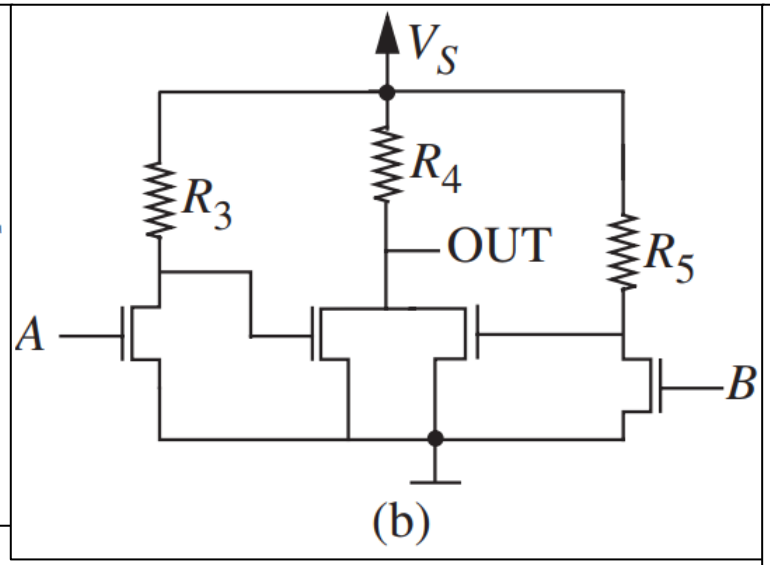
  III. SR −Model with $R_L = 40\ k\Omega$.

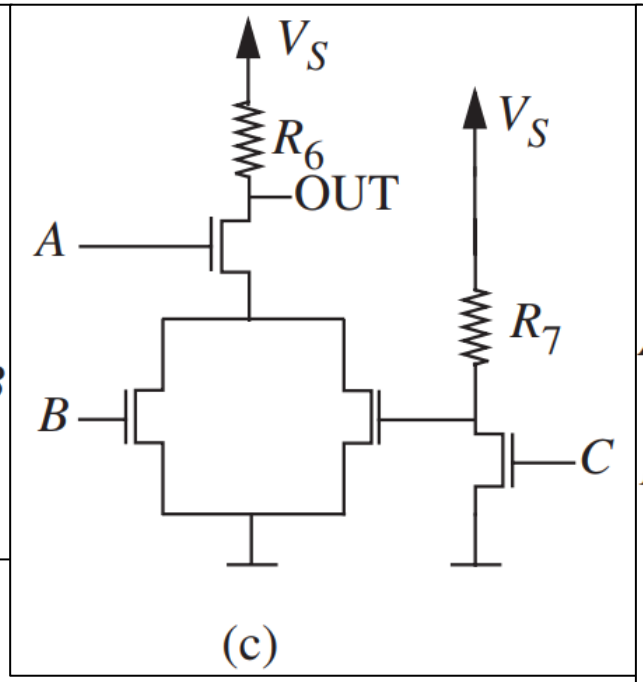


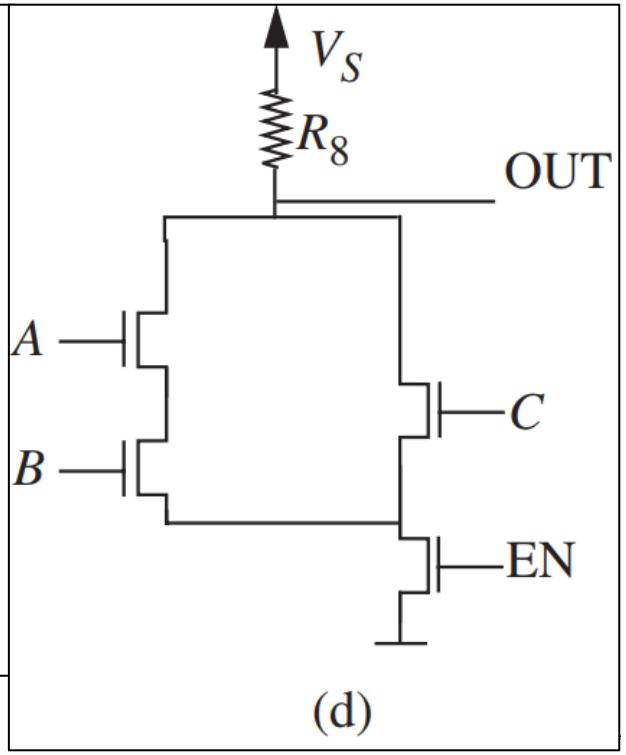
Problem 5

- The static discipline of an electronic system is such that, an input or output will be considered "low" if it remains below $0.5\,V$. Determine minimum values for the resistors R_1 through R_8 in terms of R_{ON}, so that each circuit satisfies the static discipline of the system. Here, $V_S = 5\,V$.



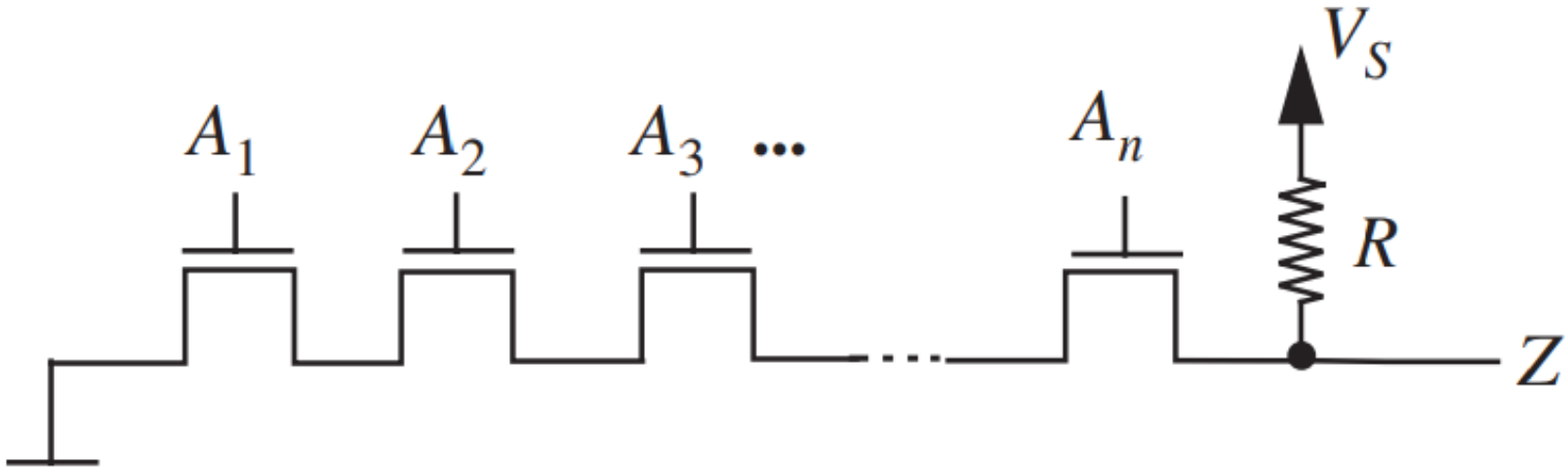






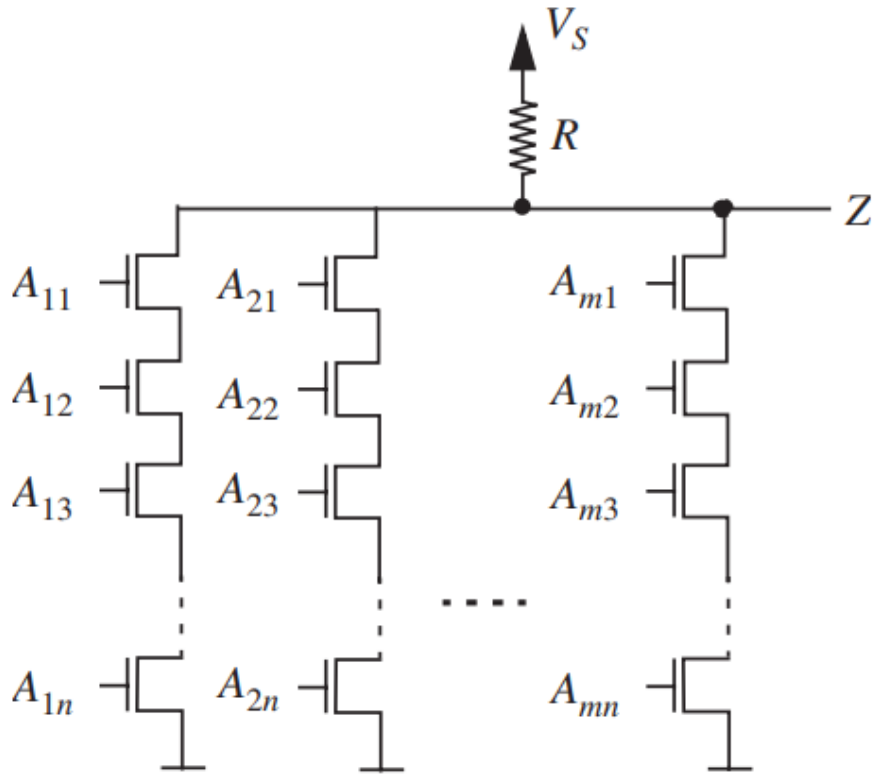
Problem 6

- The static discipline of an electronic system is such that, an input or output will be considered "low" if it remains below $0.5\,V$. Consider the N-input NAND gate circuit shown below. In the design, $V_S = 5\text{ V}$, $R = 100\,k\Omega$, and $R_{ON} = 2\,k\Omega$. Determine the maximum value of N, that is, the maximum number of MOSFETs that can be connected so that the circuit satisfies the static discipline.



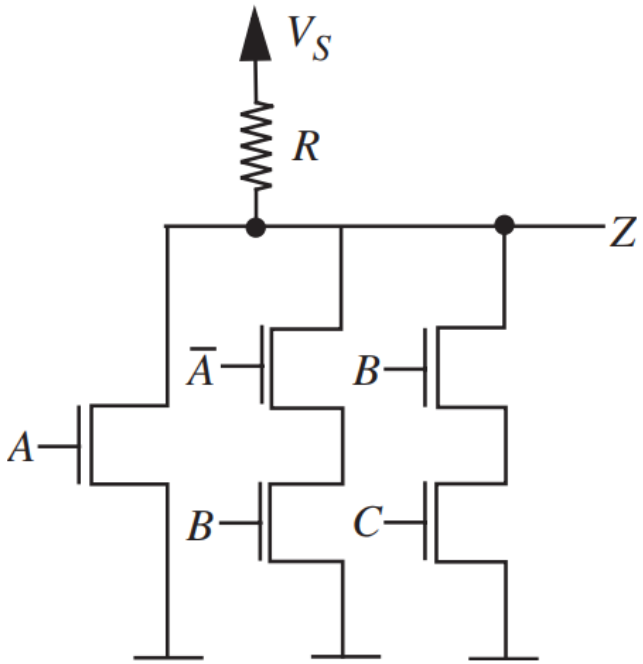
Problem 7

- The static discipline of an electronic system is such that, an input or output will be considered "low" if it remains below $0.5\ V$. Consider the following logic circuit where, $V_S = 5\ V$ and $R_{ON} = 1\ k\Omega$. Design the circuit so that the circuit satisfies the static discipline for $m = 10$ and $n = 25$.

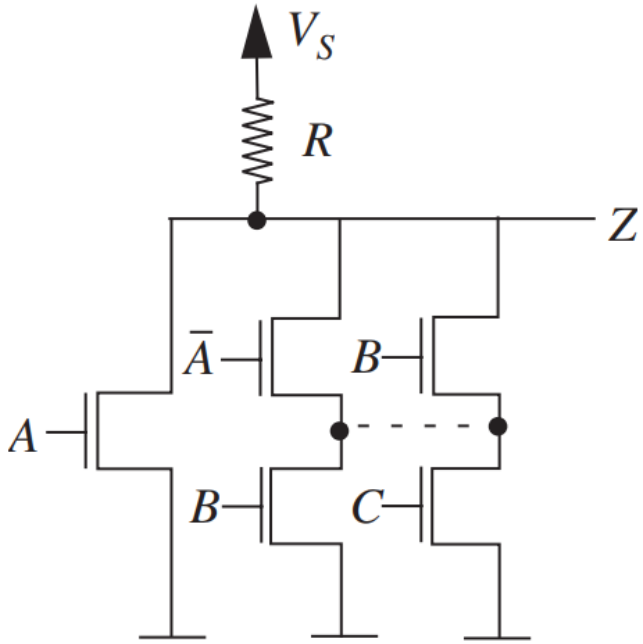


Problem 8

- Write a Boolean expression that describes the function of the circuit in figure (a). What will be the expression if a manufacturing error results in a short circuit as indicated by the dashed line in (b).



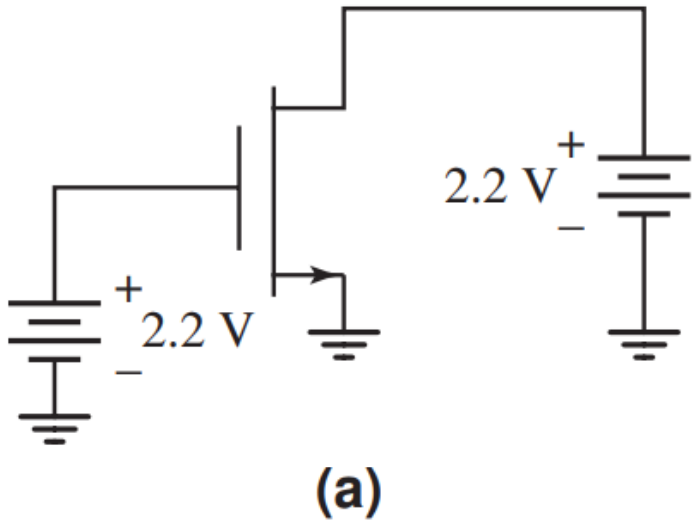
(a)



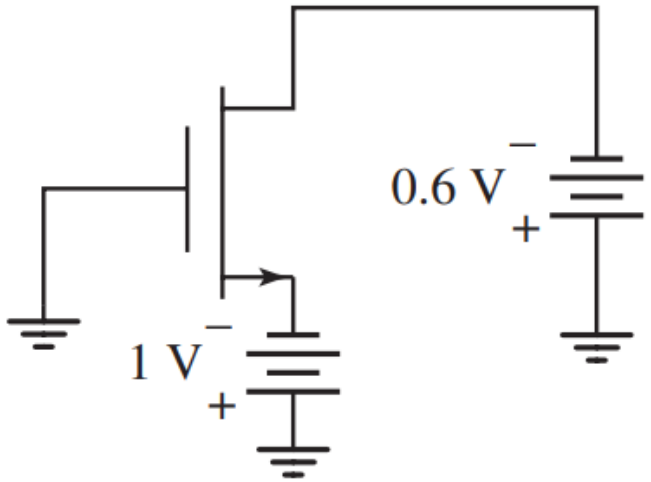
(b)

Problem 9

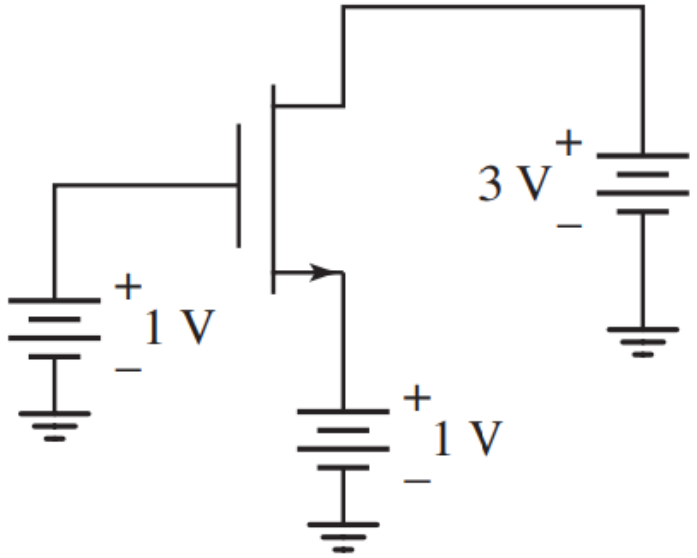
- The threshold voltage for each of the NMOS transistors in the following circuits is $V_{Tn} = 0.4\ V$. Determine the operating region of the transistor in each circuit.



(a)



(b)



(c)

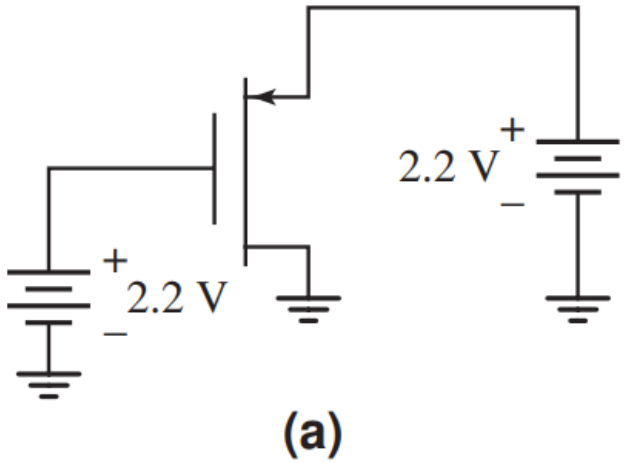
Ans: ***Saturation***

Ans: ***Triode***

Ans: ***Cutoff***

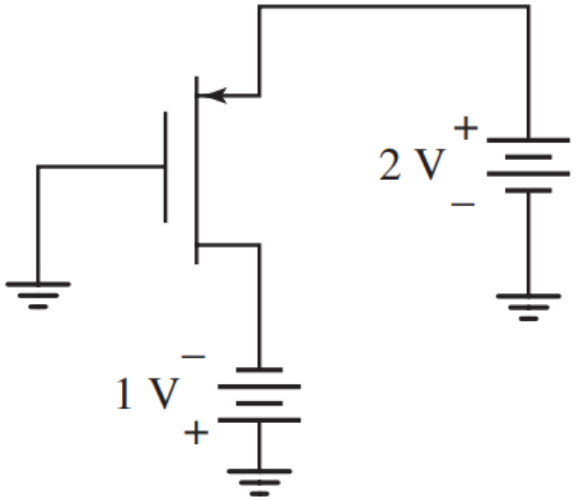
Problem 10

- The threshold voltage for each of the PMOS transistors in the following circuits is $V_{Tp} = -0.4\ V$. Determine the operating region of the transistor in each circuit.



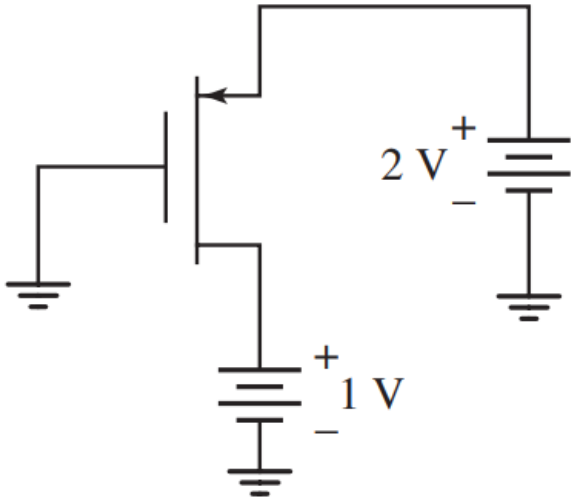
(a)

Ans: ***Saturation***



(b)

Ans: ***Saturation***

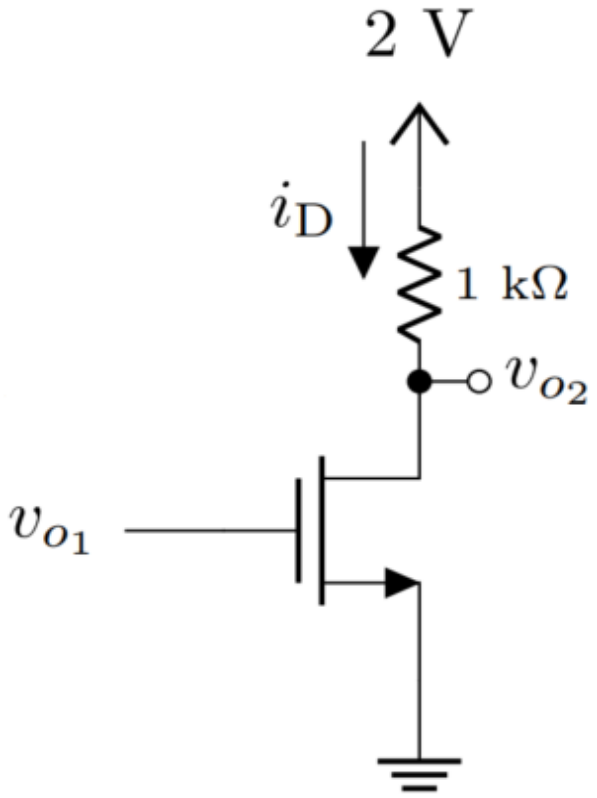


(c)

Ans: ***Triode***

Problem 11 Fall 2022

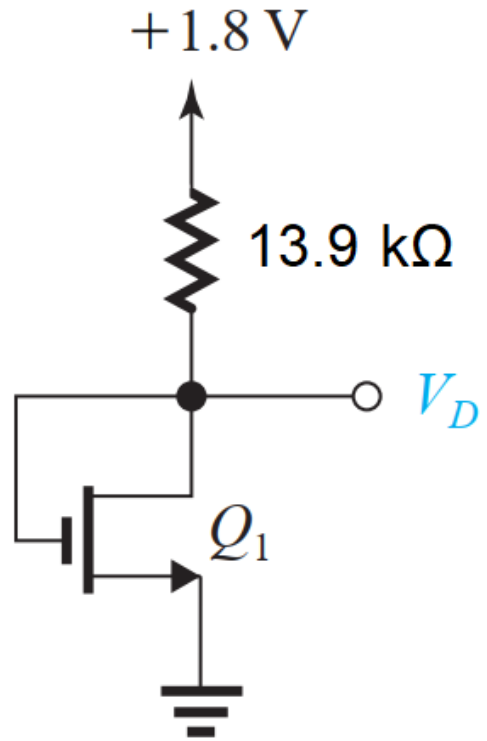
- The transistor in the following circuit has parameters $V_{Tn} = 0.2\,V$ and $k_n = k_n' \frac{W}{L} = 4\ {}^{mA}/_{V^2}$. If $v_{o_1} = 2\,V$, determine i_D and v_{o_2}.



Ans: $\boldsymbol{Triode,\ v_{o_2} = 0.26\,V; i_D = 1.74\,mA}$

Problem 12

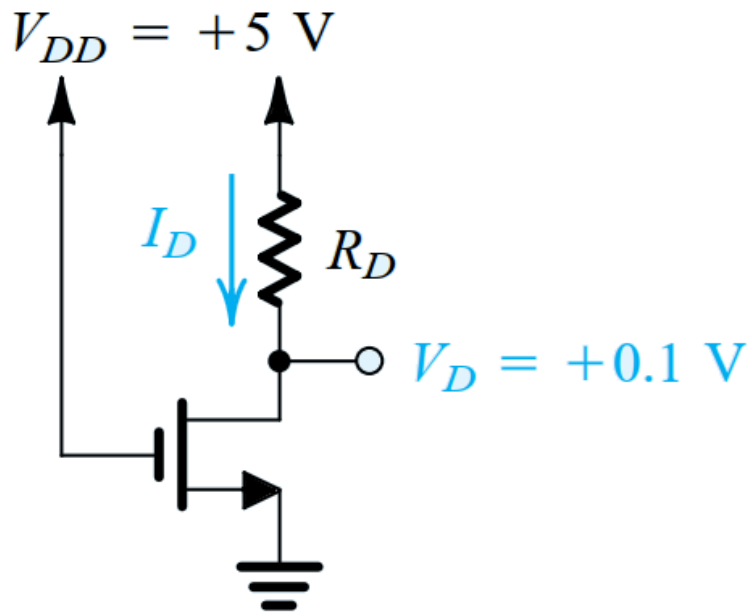
- The transistor in the following circuit has parameters $V_{Tn} = 0.5\,V$ and $k_n = k_n' \frac{W}{L} = 1.6\ {}^{mA}/_{V^2}$. Determine V_D.



Ans: $\boldsymbol{Sat,\ V_D = 0.79\,V}$

Problem 13

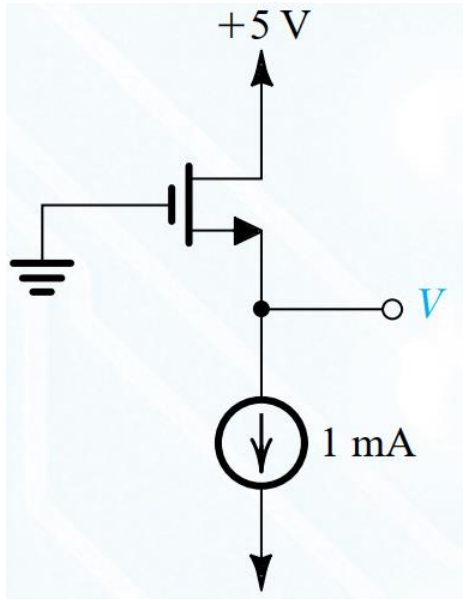
- Design the circuit, that is, determine the values of R_D, so that the transistor operates at $V_D = 0.1\,V$. The NMOS transistor has $V_{Tn} = 1\,V$ and $k_n = k_n' \frac{W}{L} = 1\ {}^{mA}/_{V^2}$.



Ans: $\boldsymbol{Triode,\ I_D = 0.395\,mA; R_D = 12.4\,k\Omega}$

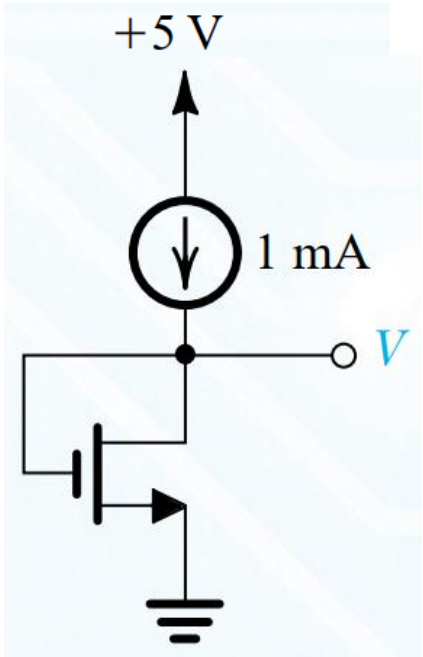
Problem 14

- The transistors in the following circuits has $V_{Tn} = 0.8\,V$ and $k_n = k_n' \frac{W}{L} = 0.5\ {}^{mA}/_{V^2}$. Determine V in each circuit.



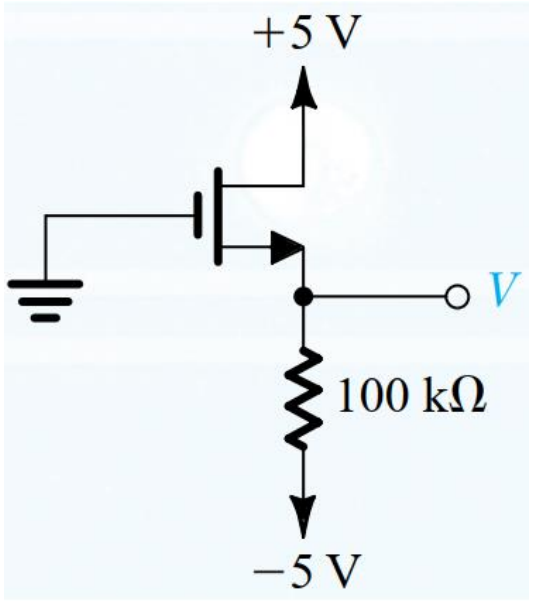
(a)

Ans: $\boldsymbol{Sat,\ \ V = -2.8\,V}$



(b)

Ans: $\boldsymbol{Sat,\ \ V = 2.8\,V}$

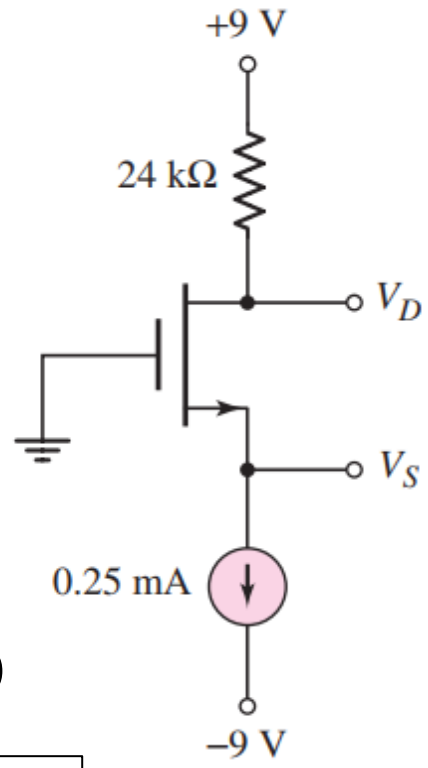


(c)

Ans: $\boldsymbol{Sat,\ V = 1.19\,V}$

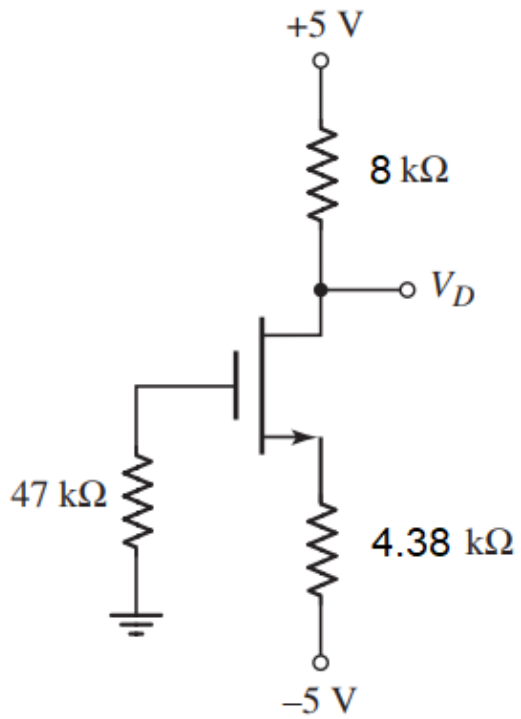
Problem 15

- The transistors in the following circuits has parameters $V_{Tn} = 0.6\,V$ and $k_n = k_n' \frac{W}{L} = 200\ {}^{\mu A}/_{V^2}$. Determine V_D and V_S.



(a)

Ans: $\boldsymbol{Sat,\ V_D = 3\,V, V_S = -2.18\,V}$

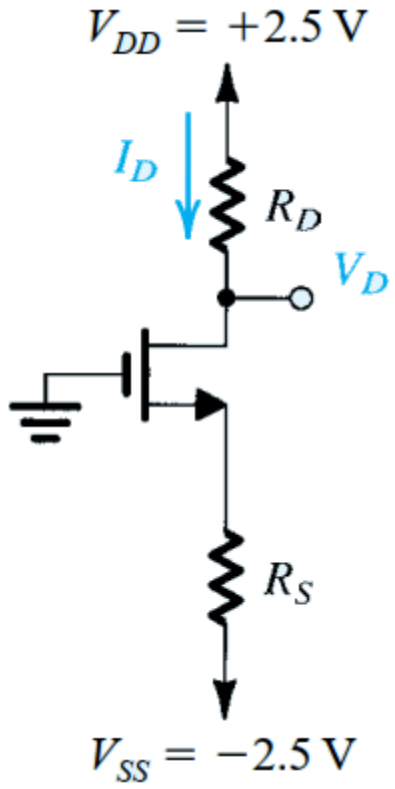


(b)

Ans: $\boldsymbol{Sat,\ V_D = 2.56\,V, V_S = -3.66\,V}$

Problem 16

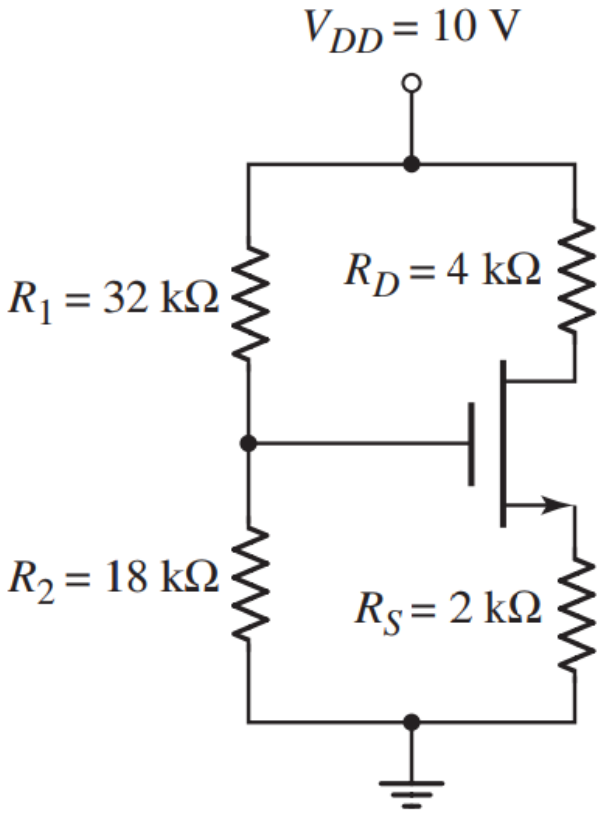
- Design the circuit, that is, determine the values of R_D and R_S, so that the transistor operates at $I_D = 0.4\,mA$ and $V_D = 0.5\,V$. The transistor has $V_{Tn} = 0.7\,V$ and $k_n = k_n' \frac{W}{L} = 3.2\ {}^{mA}/_{V^2}$.



Ans: $\boldsymbol{Sat},\ \boldsymbol{R_D = 5\,k\Omega, R_S = 3.25\,k\Omega}$

Problem 17

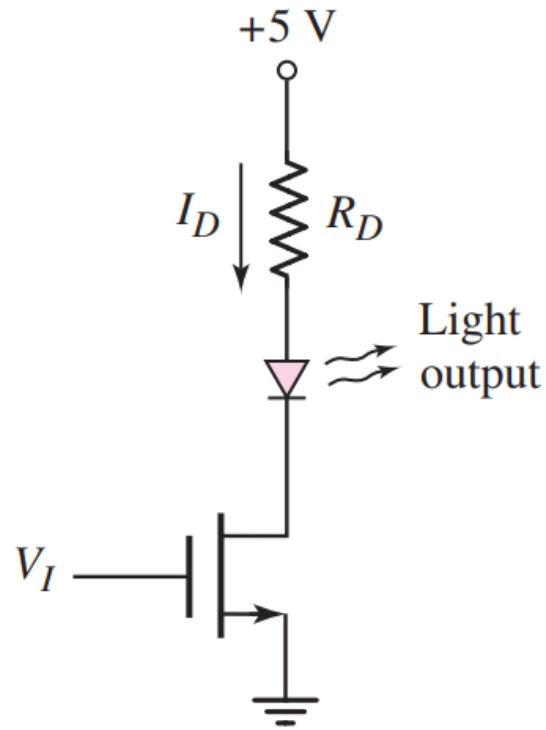
- The transistor in the following circuit has parameters $V_{Tn} = 0.8\,V$ and $k_n = k_n' \frac{W}{L} = 0.5\ {}^{mA}/_{V^2}$. Determine the voltages across the transistor.



Ans: $\boldsymbol{Sat,\ V_G = 3.6\,V, V_D = 7.54\,V, V_S = 1.23\,V}$

Problem 18

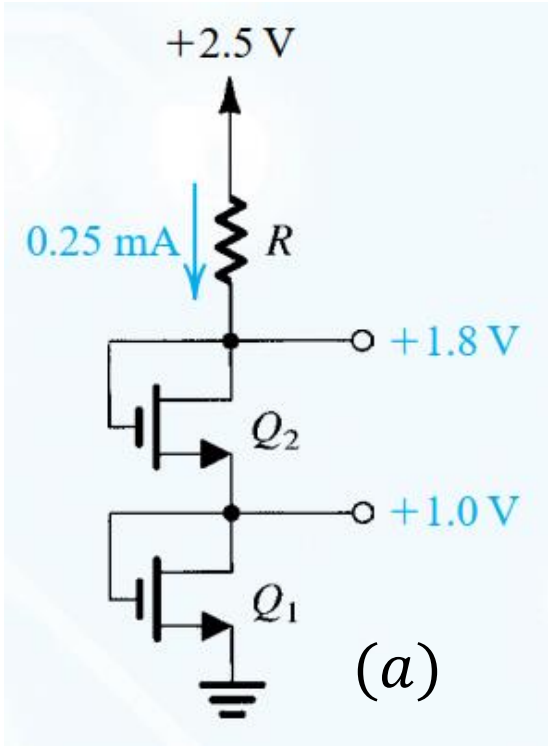
- The transistor in the following circuit is used to turn the LED on and off. The transistor parameters are $V_{Tn} = 0.6\,V$ and $k_n' = 800\ {}^{\mu A}/_{V^2}$. The diode cut-in voltage is $V_{D_o} = 1.6\,V$. Design R_D and transistor width-to-length $\left(\frac{W}{L}\right)$ ratio such that $I_D = 12\,mA$ for $V_I = 5\,V$ and $V_{DS} = 0.15\,V$.

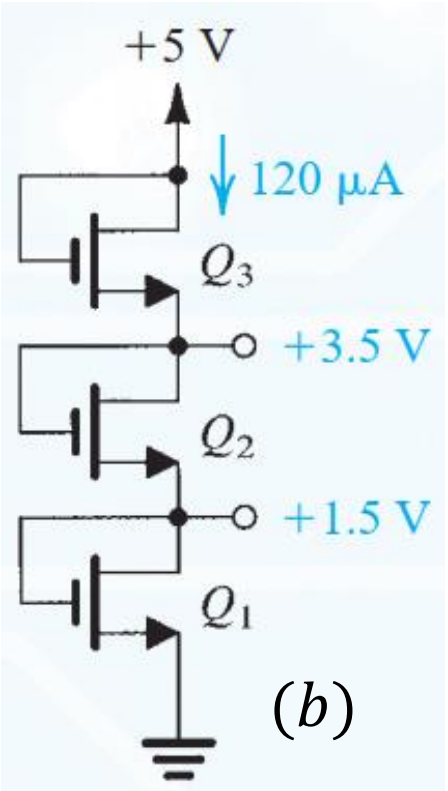


Ans: $\boldsymbol{Triode,\ R_D \approx 0.27\ k\Omega,\ \left(\frac{W}{L}\right) \approx 23}$

Problem 19

- The transistors in the following circuits has parameters $V_{Tn} = 0.5\,V$, $k_n' = \mu_n C_{oc} = 250\ {}^{\mu A}/_{V^2}$, and $L = 0.25\,\mu m$. Determine the required values of gate width for each of the transistors.



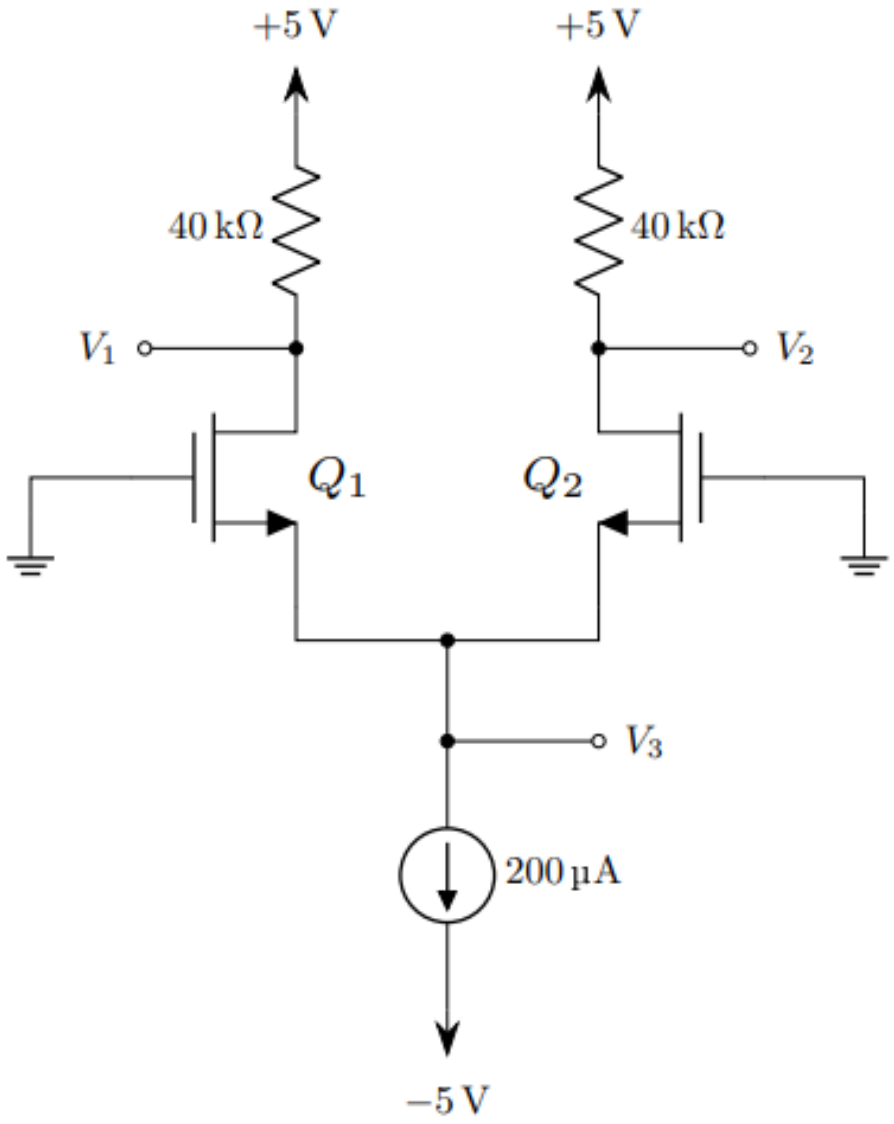


Ans: (a) $\boldsymbol{Sat,\ W_{Q_2} = 5.55\,\mu m, W_{Q_1} = 0.5\,\mu m}$; $\boldsymbol{(b)\ Sat,\ W_{Q_3} = 0.24\,\mu m, W_{Q_2} = 0.107\,\mu m, W_{Q_1} = 0.24\,\mu m}$

Problem 20

- For the identical transistors in the following circuits, $V_{Tn} = 1\,V$, $k_n' = 10\ {}^{\mu A}/_{V^2}$, and $\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2 = 20$. Determine V_1 through V_3.

[Hint: Note that both the transistors are identical and equally biased. Hence, they will have the same node voltages and conduct equal currents. Analyzing one is sufficient.]

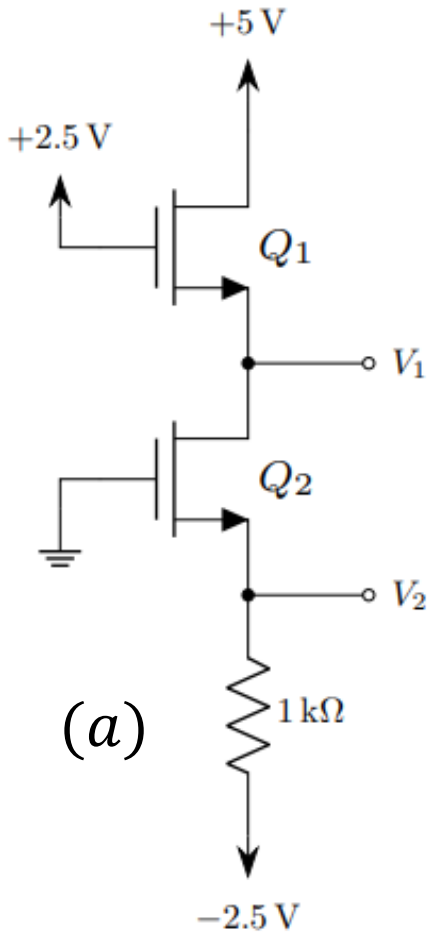


Ans: $\boldsymbol{Both\ in\ Sat,\ V_1 = V_2 = 1\,V, V_3 = -2\,V}$

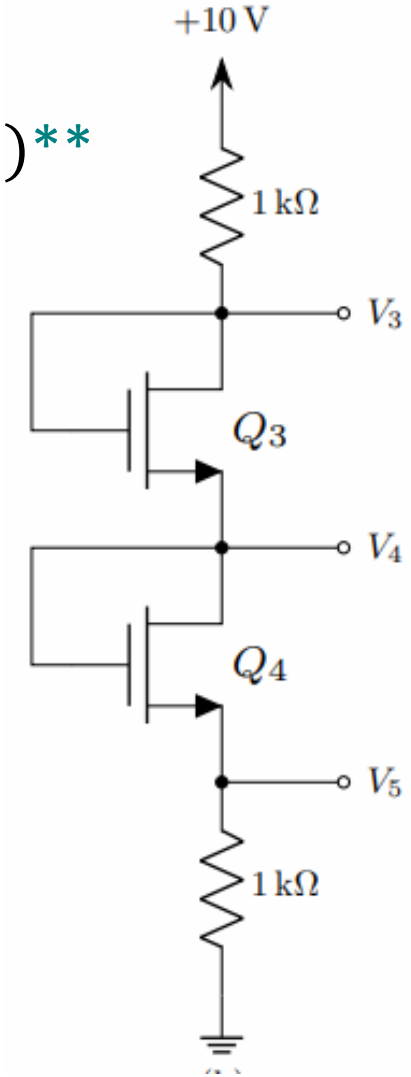
Problem 21

- For the transistors in the following circuits, $V_{Tn} = 1\ V$ and $k_n = k'_n \frac{W}{L} = 2\ {}^{mA}/_{V^2}$. Determine V_1 through V_5.

[Hint: Form simultaneous equations consisting of voltage variables for circuit in (b).]



(b)**



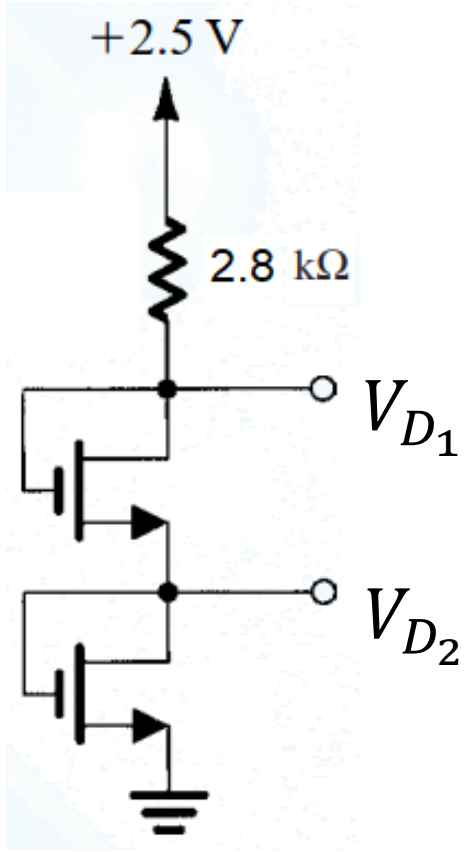
Ans: (a) $\boldsymbol{Both\ in\ sat, V_1 = 0.82\ V, V_2 = -1.82\ V}$

Ans: (b) $\boldsymbol{Both\ in\ sat, V_3 = 7.55\ V, V_4 = 5\ V, V_5 = 2.45\ V}$

Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

Problem 22**

- For the transistors in the following circuit, $V_{Tn} = 1\ V$ and $k_n = k_n' \frac{W}{L} = 5\ {}^{mA}/_{V^2}$. Determine V_{D_1} and V_{D_2}.



Ans: $\boldsymbol{Both\ in\ Sat,\ V_{D_1} = 2.72\ V, V_{D_2} = 1.36\ V}$

Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

Up Next

Relevant *Questions* from Past Semesters

Final – Fall 2025

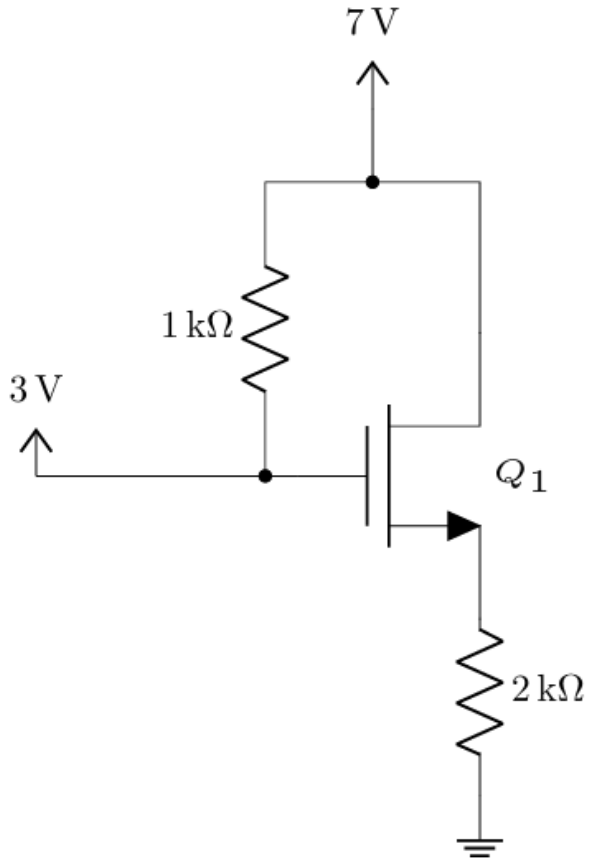
- The NMOS transistor Q_1 in the following circuit has parameters given in the box. Determine the current supplied by the 7 V source.

N-Channel MOSFET Parameters:

For the transistor Q_1,

Threshold voltage: $V_{T_n} = 2\,\text{V}$

Transconductance parameter: $k_n = 2\,\text{mA/V}^2$



Ans: $(a)\ \boldsymbol{Q_1 \rightarrow Sat;\ V_S = 2\,V; I_{7\,V} = 4.25\,mA}$

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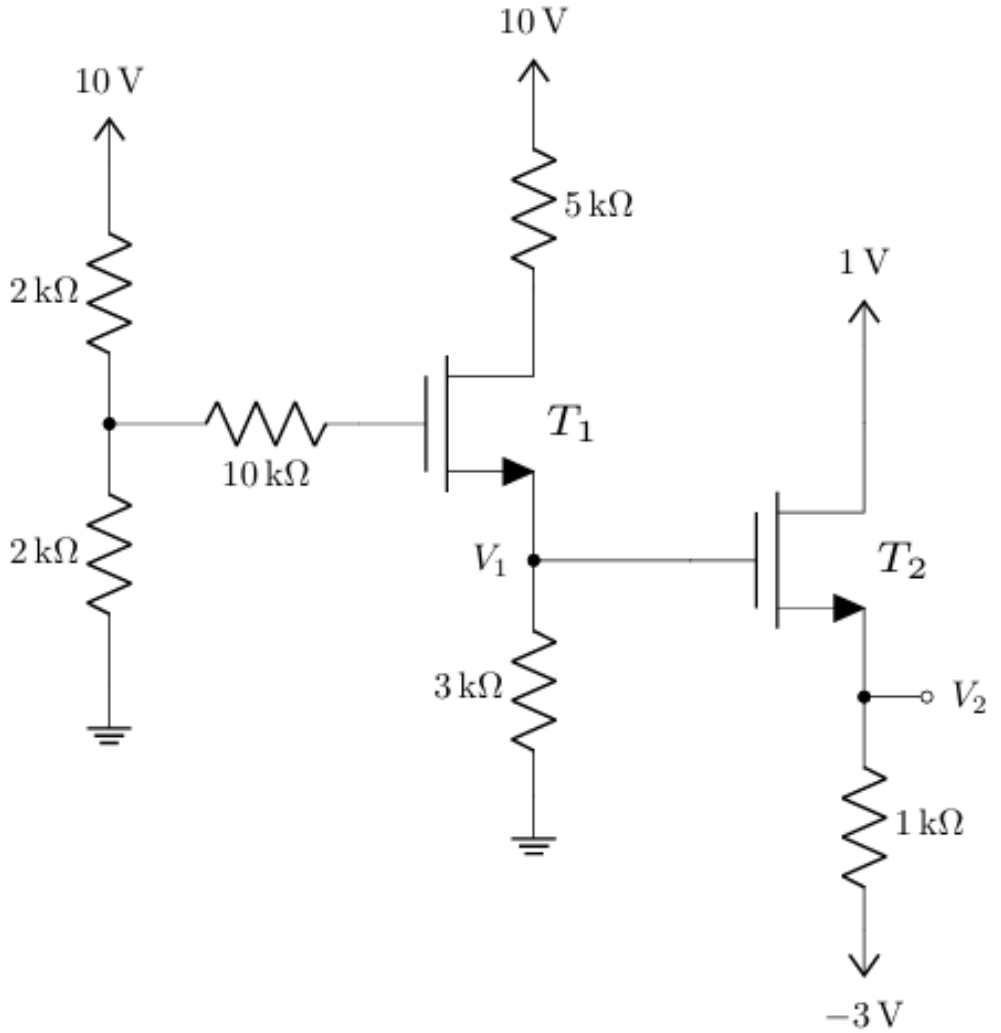
- The transistors T_1 and T_2 in the following circuit are identical and have the parameters given in the box. If the transistor T_1 operates in the saturation region,
  a. Determine V_1 and the drain voltage of T_1.
  b. Using the value of V_1 calculated in (a), determine the voltage V_2 by verifying the operating region of T_2.

MOSFET Parameters:

For both T_1 and T_2:

Threshold voltage: $V_{T_n} = 1\,\text{V}$

Transconductance parameter: $k_n = 2\,\text{mA/V}^2$



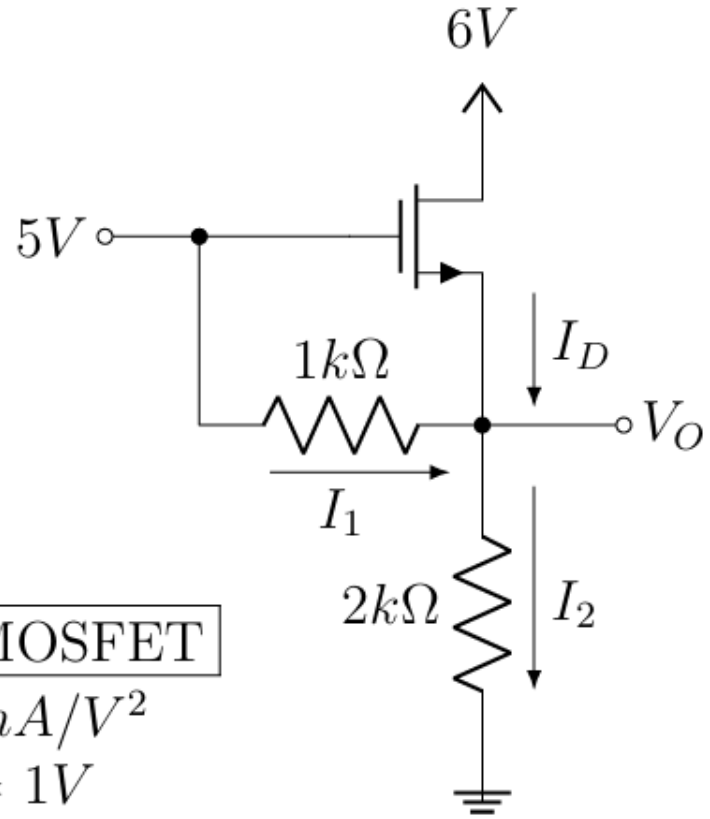
Ans: $(a)\ \boldsymbol{T_1 \rightarrow Sat;\ V_{G_1} = 5\,V; V_{D_1} = 5\,V; V_1 = V_{S_1} = 3\,V; I_{DS_{T_1}} = 1\,mA}$

$(b)\ \boldsymbol{T_2 \rightarrow Triode; V_2 = V_{S_2} = 0\,V; I_{DS_{T_2}} = 3\,mA}$

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a. Analyze the adjacent circuit, and show that, the MOSFET will operate in Saturation mode if it conducts current. *[Hint: You don't need to solve the circuit to proof that.]*

b. Calculate V_O, I_1, I_2, and I_D using the method of assumed states. You must validate your assumption.



For the MOSFET

$k = 1mA/V^2$

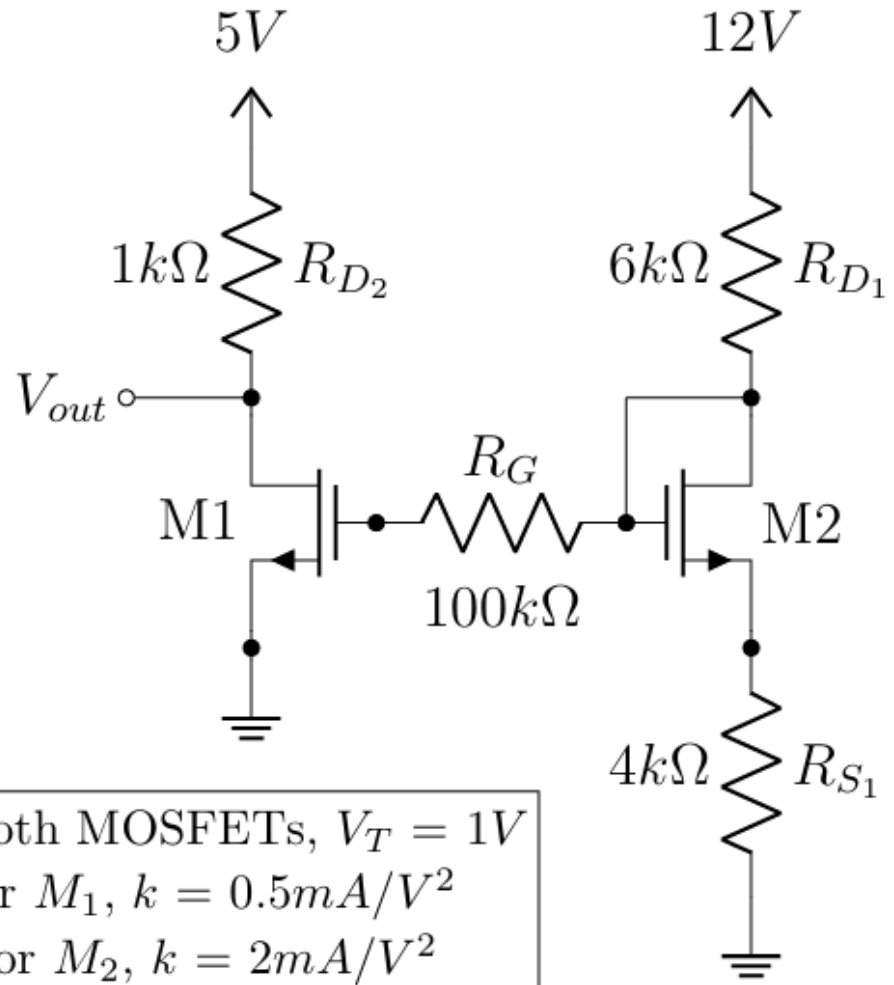
$V_T = 1V$

Ans: $(b)\ V_O = 3.63\ V; I_1 = 1.57\ mA; I_2 = 1.72\ mA; I_{DS} = I_D = 0.15\ mA$

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a. Show that, if the MOSFET, M2 conducts current, it will operate in the saturation mode. *[Hint: You don't need to solve the circuit to proof that.]*

b. Calculate V_{out}, $I_{R_{D_1}}$, and $I_{R_{D_2}}$ using the method of assumed states. You must validate your assumption.



for both MOSFETs, $V_T = 1V$
for M_1, $k = 0.5mA/V^2$
for M_2, $k = 2mA/V^2$

Ans: (b) $\mathbf{M2} \rightarrow \boldsymbol{Sat}$; $\boldsymbol{V_S = 4\,V; V_G = 6\,V; I_{DS} = I_{R_{D_2}} = 1\,mA}$
$\mathbf{M1} \rightarrow \boldsymbol{Triode}$; $\boldsymbol{V_{out} = 1.615\,V; I_{DS} = I_{R_{D_1}} = 3.385\,mA}$

Final – Summer 2024 - 1/2

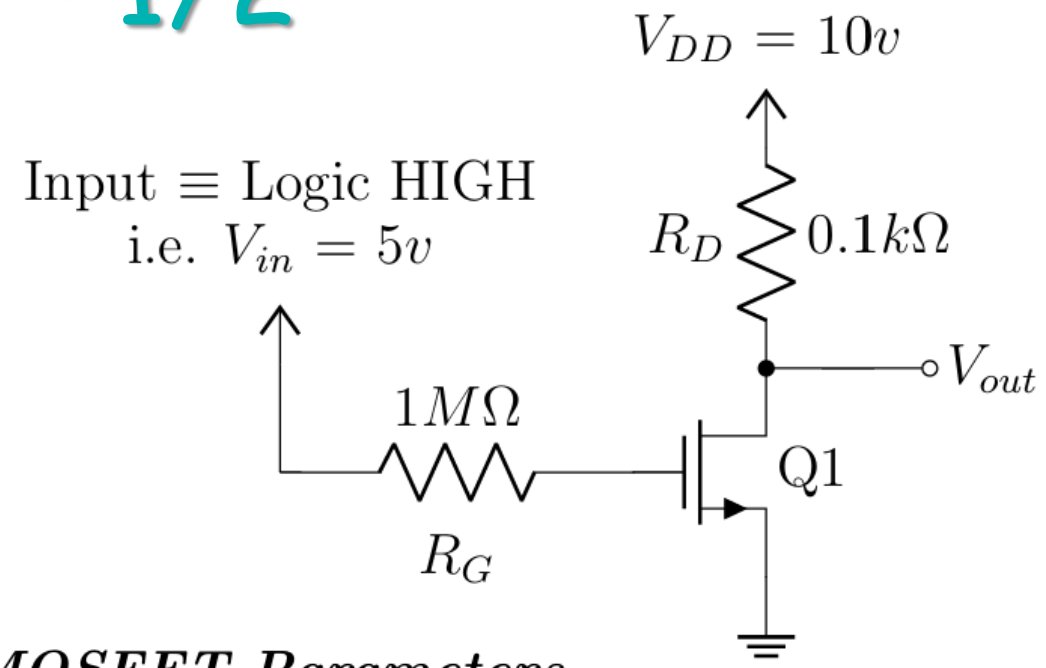
- Joy and Nirmol were designing an inverter circuit with MOSFET for their project. Joy designed the following circuit. However, Nirmol said that the designed inverter circuit would malfunction. When they tested the circuit, it malfunctioned. For logic HIGH input, the output was also logic HIGH instead of logic LOW. Nirmol modified the value of R_D to a larger value, and eventually it fixed the issue.

a. Calculate the value of V_{out} using the method of assumed states. You must validate your assumption. Determine whether Nirmol was correct or not.

b. Nirmol changed the value of R_D to $1\ k\Omega$. Determine whether the the inverter was working properly after the modification.

c. Design the inverter circuit by choosing a suitable value of R_D such that, the MOSFET operates at the edge of saturation.

Ans: (a) $\boldsymbol{Sat;\ V_{out} = 6.8\ V; I_{DS} = 32\ mA}$
(b) $\boldsymbol{Triode;\ V_{out} = 0.636\ V; I_{DS} = 9.364\ mA}$
(c) $\boldsymbol{R_D = 0.1875\ k\Omega}$



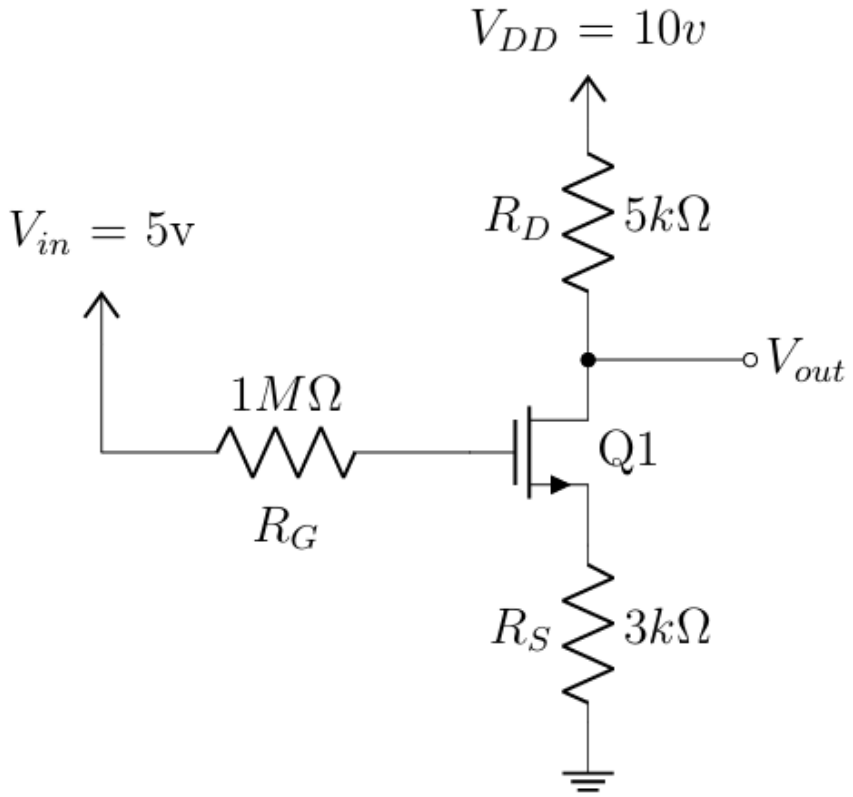
MOSFET Parameters

$V_T = 1v,\ K = 4mA/v^2$

For Input		For Output	
Voltage Level	*Logic Level*	*Voltage Level*	*Logic Level*
$V_{in} = 0v$	LOW i.e. '0'	$V_{out} > 5v$	HIGH i.e. '1'
$V_{in} = 5v$	HIGH i.e. '1'	$V_{out} < 5v$	LOW i.e. '0'

Final – Summer 2024 - 2/2

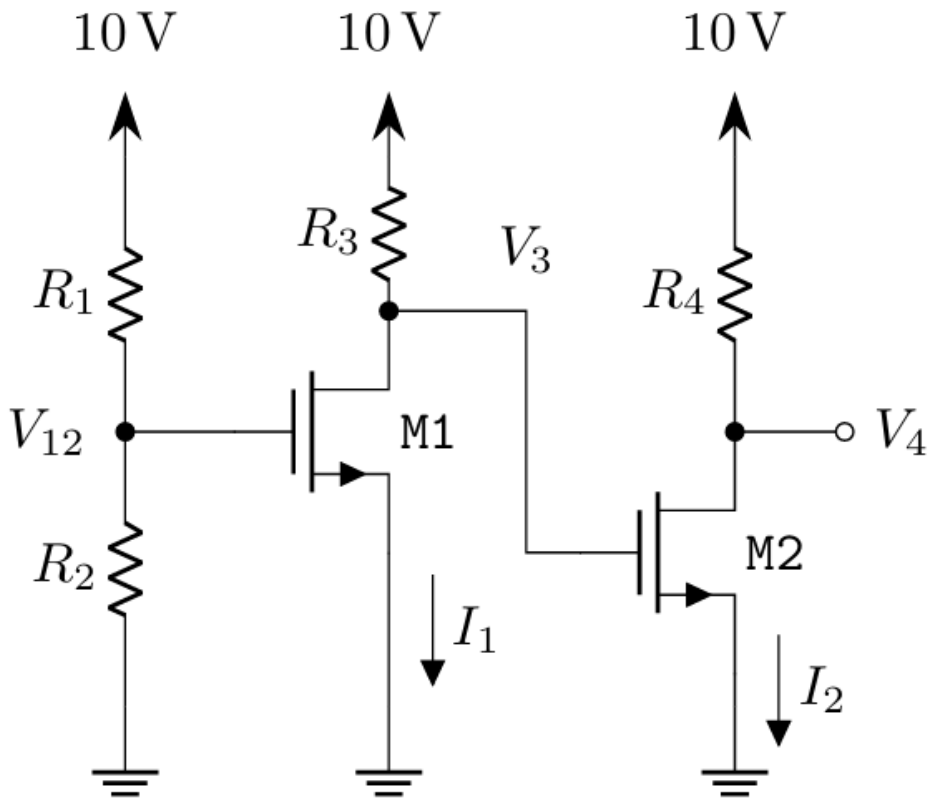
- Analyze the circuit and calculate V_{out} using method of assumed states. You must validate your assumptions. For the transistor, $V_{T_n} = 1\ v$ and $k_n = 2\ {}^{mA}/_{v^2}$.



Ans: $\boldsymbol{Sat;\ V_{out} = 5\ V; V_S = 3\ V; I_{DS} = 1\ mA}$

Final – Spring 2024

- For the adjacent circuit,
  a. Determine V_{12}.
  b. Calculate the value of drain-to-source voltage of the transistor M1. In the process, determine the operating mode of M1.
  c. Determine which one of the four resistors will consume the least amount of power.
  d. Determine how the operating mode of M2 would change if R_2 were set to zero.



Resistor values in (kΩ)

R_1	R_2	R_3	R_4
3	5	2	4

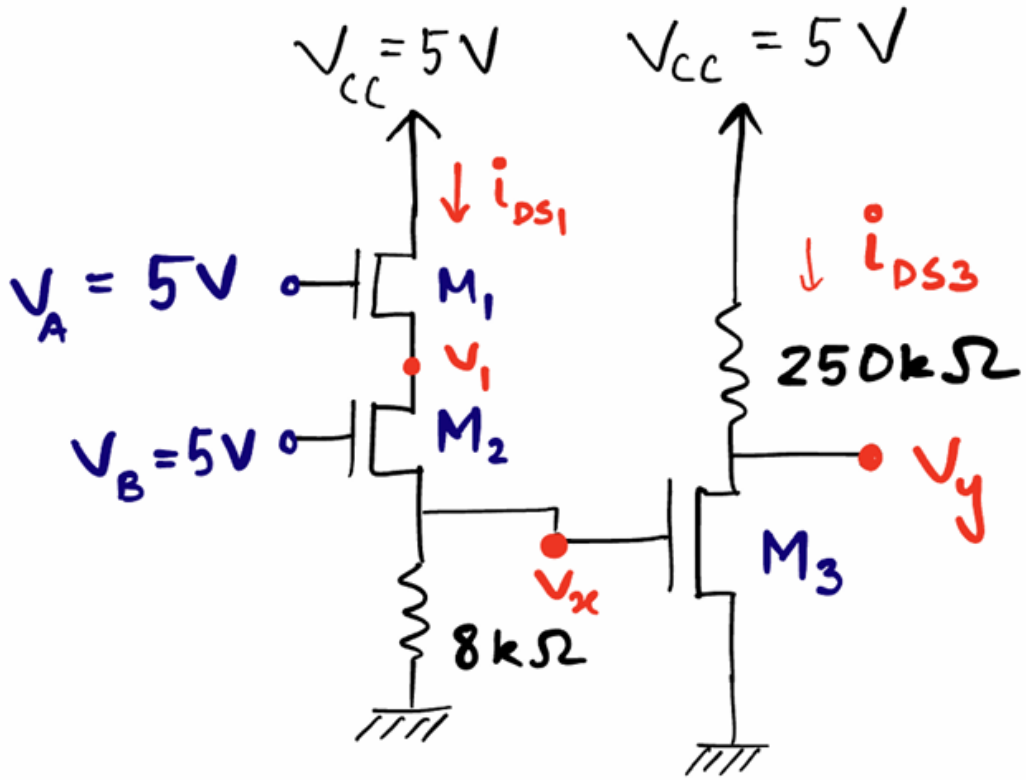
Transistor Parameters

$k_n = 2\ \text{mA/V}^2$
$V_T = 1\ \text{V}$

Ans: (a, b) **M1** $\rightarrow$ ***Triode***; $V_{12} = V_{G_{M1}} = 6.25\ V; V_3 = V_{D_{M1}} = V_{DS_{M1}} = V_{G_{M2}} = 0.475\ V; I_{DS} = I_1 = 4.76\ mA$
(c) **M2** $\rightarrow$ ***Cutoff***; $V_4 = V_{D_{M2}} = V_{DS_{M2}} = 10\ V; I_{DS} = I_2 = 0\ mA; P_{min} = P_{R_4} = 0\ mW$
(d) **M1** $\rightarrow$ ***Cutoff***; $V_{12} = V_{G_{M1}} = 0\ V; V_3 = 10\ V$; ***M2*** $\rightarrow$ ***Triode***; $V_4 = V_{D_{M2}} = V_{DS_{M2}} = 0.138\ V$

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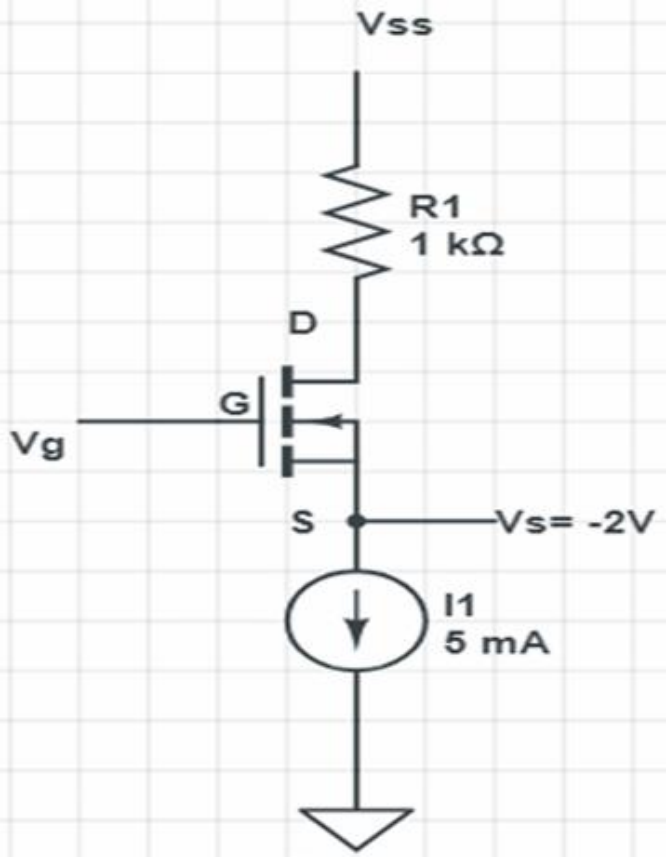
- For the MOSFETs in the adjacent circuit, threshold voltage $V_{T_n} = 1\,V$, process transconductance parameter $k_n' = 50\ {}^{\mu A}/_{V^2}$, and aspect ratio ${}^{W}/_{L} = 1$.
  a. If the gate voltage to M_3 is $V_x = 0.938\,V$, and M_2 is in triode mode, find the voltage V_1.
  b. Determine the value of V_y.
  c. If ${}^{W}/_{L} = 2$, then $V_1 = 2.1454\,V$. Find V_x for this case.
  d. Will there be any changes in the output voltage at V_y for ${}^{W}/_{L} = 2$ from that in (b)? Explain briefly. Also, comment on whether a larger or smaller value of ${}^{W}/_{L}$ ratio is preferred in this case.



Ans: $(a)\ \mathbf{M_1} \rightarrow \boldsymbol{Sat};\ \boldsymbol{V_1 = 1.83\,V}$
$(b)\ \mathbf{M_3} \rightarrow \boldsymbol{Cutoff};\ \boldsymbol{V_y = 5\,V}$
$(c)\ \boldsymbol{V_x = 1.375\,V}$
$(d)\ \mathbf{M_3} \rightarrow \boldsymbol{Sat};\ \boldsymbol{V_y = 3.24\,V};\ \boldsymbol{Large\ {}^{W}/_{L}\ preferred}$

Final – Fall 2023 -2/3

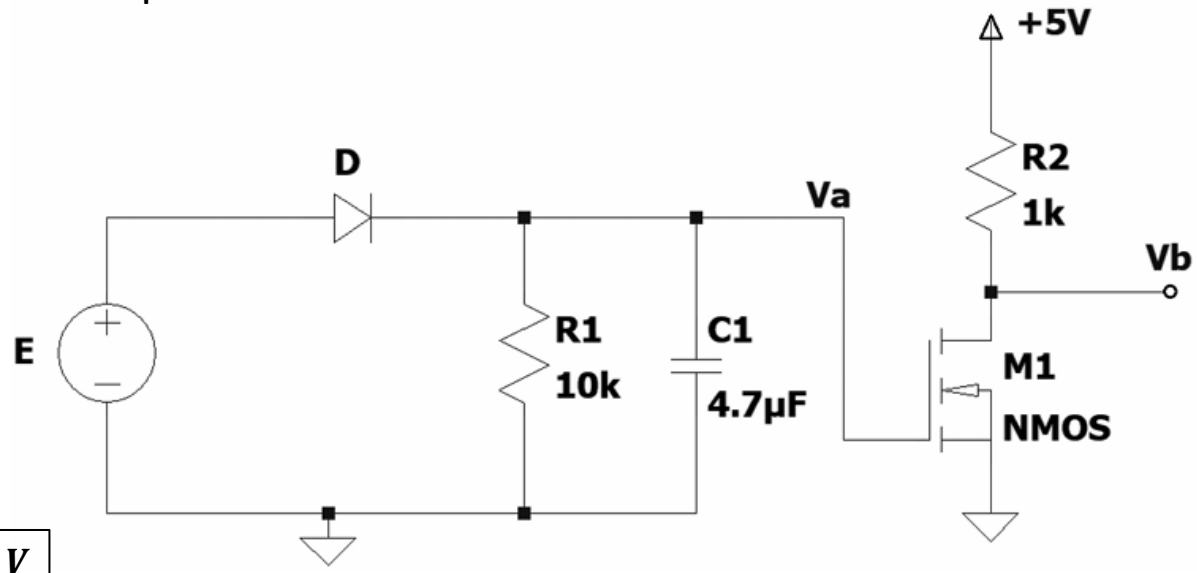
- For the MOSFET in the adjacent circuit, $V_{T_n} = 1\,V$ and $k_n = 2\ {}^{mA}/_{V^2}$. Determine the gate voltage V_g and the minimum supply voltage V_{ss} required to push the transistor to operate in saturation.



Ans: $\boldsymbol{Edge\ of\ Sat;\ V_g = 1.236\,V; V_D = 0.236\,V; V_{SS} = 5.236\,V}$

Final – Fall 2023 -3/3

- A Half-wave rectifier circuit has been cascaded with an inverter as shown in the following circuit. You are given the following information about the circuit: $E = 5\sin(400\pi t)\ V$, $k_n = 2\ {}^{mA}/_{V^2}$, $V_{T_n} = 1\ V$, and $V_{D_0} = 0.7\ V$.
  a. Determine the peak-to-peak ripple value of the half-wave rectifier output Va. What is the lowest value of Va?
  b. At the lowest value of Va, determine the output of the inverter Vb.
  c. Now disconnect the capacitor. In this case, find out the voltage Vb for the lowest value of Va.
  d. Draw an approximate plot of Vb without the capacitor connection.



Ans: $(a)\ Va = 0.5\ V,\ Va_{Lowest} = 4.45\ V; (b)\ Vb = 0.69\ V; (c)\ Vb = 5\ V$

Final – Spring 2023

a. Suppose you have 3 MOSFETs labeled A, B, and C that you want to use as switches. You want to connect them together in a circuit so that, the output is ON only when switch A is ON and either switch B or switch C is also ON. Using the given MOSFETs, design and draw a circuit that gives the desired output. You can use other necessary components like resistors, but you can only use the 3 MOSFETS provided.

b. You have a MOSFET inverter with a supply voltage $V_S = 5\,V$. The MOSFET has parameters: $k'_n = 1\frac{mA}{V^2}$, $\frac{\mathrm{W}}{\mathrm{L}} = 4$, and $V_{T_n} = 0.6\,V$. Determine the on-state resistance R_{ON} for the transistor when the input voltage is $5\,V$. Assume that the transistor is biased deep into Triode.

Ans: $(\boldsymbol{b})\ \boldsymbol{R_{ON} = 56.82\ k\Omega}$

Final – Fall 2022

- In digital systems, binary data may be subjected to noise that can alter a 0 to a 1 or a 1 to a 0. A simple way to check if any error has occurred is to use an even parity checker. If there are two input bits x and y, the output of the even parity checker (denoted as f) will be HIGH if there are even number of $1s$, i.e., if both x and y are 0 or if both of them are 1.
  a. Deduce the logic function for f in terms of the Boolean inputs x and y.
  b. Design a circuit using MOSFET logic gates to implement this function.

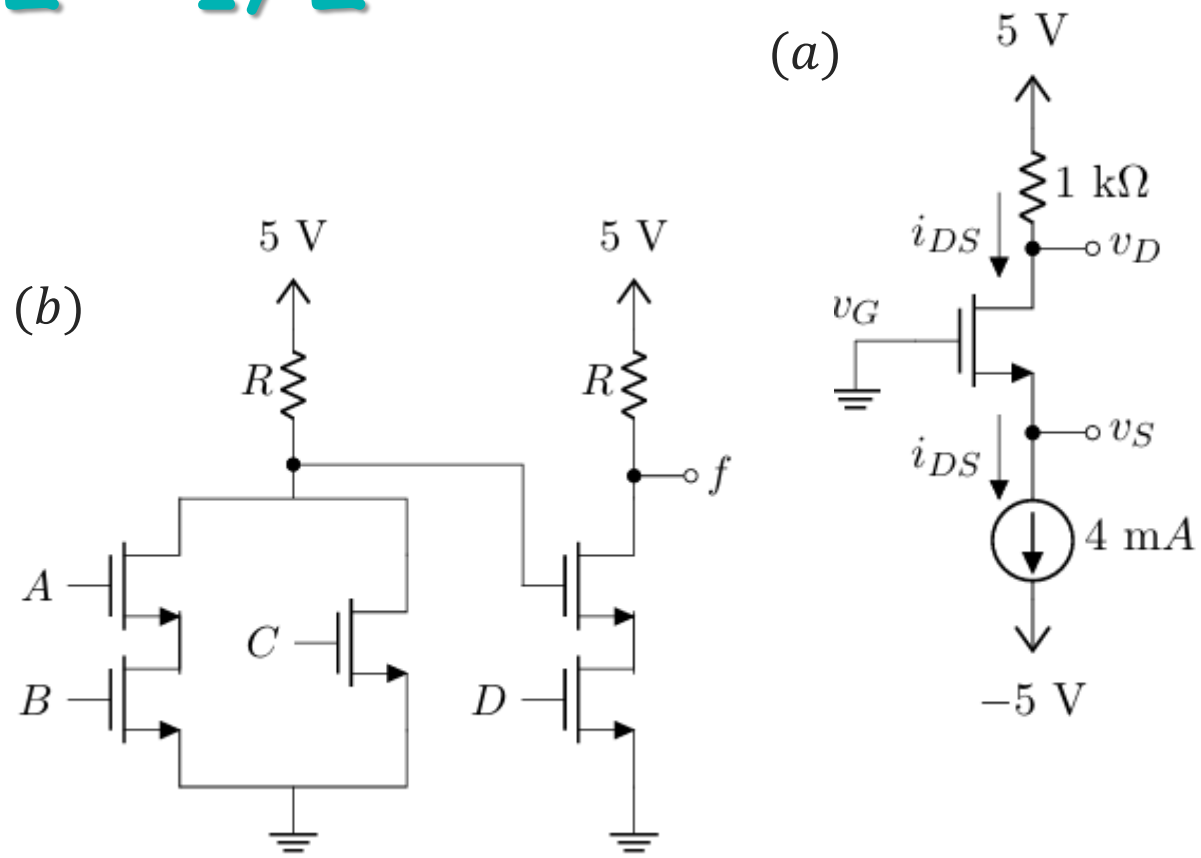
Ans: $(a)\ f = x.y + \overline{x}.\overline{y}\ \ or\ f = \overline{\overline{x}.y + x.\overline{y}}$

Final – Summer 2022 - 1/2

a. For the MOSFET in the circuit (a), $V_{T_n} = 1\,V$ and $k_n = 4\ {}^{mA}/_{V^2}$. Determine the drain voltage v_D and the source voltage v_S.

b. For the logic circuit (b), write an expression for the Boolean output f in terms of the Boolean inputs A, B, C, and D.

c. Consider a MOSFET inverter (SR model) with the following circuit parameters: supply voltage $V_S = 5\,V$, pull-up resistance $R_L = 10\,k\Omega$. Also, for the MOSFET, $V_{T_n} = 1\,V$ and $1/k_n'(V_{GS} - V_{T_n}) = 5$. Determine a W/L sizing for the MOSFET so that the inverter gate output of logical 0 is able to switch OFF the MOSFET of another inverter. Assume that the transistor is biased deep into Triode.



Ans: (a) $\boldsymbol{Triode; v_D = 1\,V; v_S = -3.45\,V;}$
(b) $\boldsymbol{R_{ON} = 2.5\,k\Omega,}$ $\boldsymbol{\frac{W}{L} = 2}$

Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Agarwal, A. and Lang, J., Foundations of Analog and Digital Electronic Circuits, Morgan Kaufmann
2. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
3. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill

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